

# Frequency-Gated Prompting for Enhancing Transformer-based EEG Decoding

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**Abstract**—The fundamental problem in electroencephalogram (EEG) decoding centers on the extraction of meaningful neural patterns from complex spatio-temporal signals. Although Transformer models have garnered significant attention in this domain due to their exceptional temporal modeling capabilities, their lack of perception for critical frequency-domain features within EEG signals constrains further performance enhancement. To address this issue, this paper proposes a lightweight approach termed frequency-gated prompted Transformer (FGPT). FGPT adaptively represents global EEG rhythms by introducing learnable sparse frequency prompt tokens. Utilizing a gated fusion mechanism, it synergistically embeds these tokens with the original EEG sequence into the Transformer's self-attention computation. This enables joint modeling of spatio-temporal and frequency-domain features without compromising sequence continuity. Experiments conducted on three public EEG datasets show that FGPT improves the decoding performance and robustness of the evaluated Transformer-based models (EEG-ViT, EEG-Conformer, and EEG-Deformer) on the selected baselines. With its lightweight design and observed potential for generalization on the datasets used, this approach explores a prompt learning method that may contribute to developing more efficient EEG decoding systems. The code can be obtained from <https://github.com/liangjiaxiaoqi/FGPT>.

**Index Terms**—EEG; Frequency prompt; Sparse token; Gate fusion; Transformer

## I. INTRODUCTION

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ELECTROENCEPHALOGRAPH (EEG) is a vital technique for non-invasively recording the electrophysiological activity of brain neurons via sensors placed on the scalp surface. The signals it captures reflect the synchronized discharge activity of large clusters of cortical neurons and possess millisecond-level temporal resolution. In scientific research and clinical practice, EEG signals undergo a series of complex processing and analysis steps, which include feature extraction and pattern recognition, to decode underlying neural activity patterns and cognitive states. This technique enables direct and entirely non-invasive monitoring of the brain's bioelectrical activity, providing a crucial window into understanding neural mechanisms. Currently, EEG technology is extensively applied in domains such as rehabilitation training [1] [2], disease treatment [3] [4], and augmented and virtual reality [5] [6].

Deep learning has achieved breakthrough progress in automatic feature extraction, significantly advancing research in electroencephalogram (EEG) decoding. As a bioelectrical sequence with dual spatio-temporal characteristics, EEG exhibits spatial dimensions that reflect collaborative relationships between different brain regions, while its temporal dimension encodes the millisecond-scale dynamic evolution of neural activity. With its global perception and temporal modeling capabilities, Transformer has become the mainstream architecture for EEG decoding. Its self-attention mechanism can directly capture long-range spatial dependencies and complex time-varying patterns across electrodes, establishing its core position in EEG classification [7]–[10]. For different decoding tasks, researchers have proposed various improved architectures. Deny et al. [11] designed a hierarchical Transformer to focus on interference-free period signals in motor imagery. Xie et al. [12] combined CNN with Transformer to learn local spatial features and global dependencies respectively, and this CNN-Transformer framework has been widely adopted [12]–[15]. Other innovations include Pfeffer et al. [16] who utilized genetic algorithms for architecture search, Liu et al. [17] who proposed a cross-subject generalization model DG-Conformer, and Wang et al. [18] who designed the pre-trained model EEGPT. In terms of module fusion, Yin et al. [19] used parallel CNN and Transformer blocks to extract local and global temporal features respectively, while Ouyang et al. [20] introduced self-attention mechanisms into convolutional modules. Additionally, Chen et al. [21] focused on EEG denoising, Lian et al. [22] integrated graph neural networks with Transformer, and Fan et al. [23] and Li et al. [24] explored multi-

task learning and time-frequency-space dual-branch feature extraction respectively. However, most existing Transformer models primarily focus on spatio-temporal information and fail to adequately model cross-domain joint representations. As a purely spatio-temporal sequence, EEG lacks explicit perception of frequency attributes, limiting its ability to analyze rhythmic neural activity.

Frequency-domain information is crucial for EEG decoding and classification, as it clearly separates different physiological and noise components in EEG recordings, thereby facilitating the revelation of brain rhythmic activities. Liu et al. [25] and Hamidi et al. [26] have enriched the feature space by introducing frequency-domain information. To address the challenge of effectively integrating multi-band features, Wang et al. [27] achieved cross-frequency feature fusion through a multi-band fusion algorithm based on Riemannian geometry. Considering individual differences and the non-stationary nature of EEG signals, Wang et al. [28] employed two time-frequency transformation-based EEG data augmentation methods, discrete wavelet transform augmentation and Hilbert-Huang transform augmentation, significantly improving the practicality of classification models in real-world applications. Liu et al. [29] utilized multi-scale time-frequency segmentation and spatial segmentation schemes to process multi-channel MI-EEG recordings, generating time-frequency-spatial segments and classifying them through feature selection methods. Jin et al. [30] extracted EEG features from multiple spatio-temporal frequencies, processing high, medium, and low frequency components in different branches and enabling interactive feature learning. This parallel multi-branch approach often incurs exponential computational overhead and structural redundancy. Additionally, Bunterngchit et al. [31] and Cai et al. [32] converted raw EEG recordings into two-dimensional spectral images via a short-time Fourier transform, then employed transfer learning or designed the corresponding models to extract discriminative features. Wang et al. [33] designed an Interactive Frequency Convolutional Neural Network (IFNet), Garg et al. [34] adopted continuous wavelet transform to convert EEG signals into time-frequency maps, Miao et al. [35] used group sparse regression to select optimal frequency bands, while Luo [36] addressed the challenges of limited EEG sample size and low signal-to-noise ratio by selectively extracting time-frequency features from multiple spatial perspectives. However, these methods typically rely on complex network structures, which not only lead to significant computational redundancy and implementation complexity, but also disrupt the continuity of EEG sequence structure.

Consequently, EEG decoding research faces a core challenge: integrating frequency-domain perception into the Transformer architecture in a lightweight manner. The ultimate goal is to achieve efficient joint feature extraction across both spatiotemporal and frequency domains. Given that EEG signals exhibit distinct time-frequency distribution characteristics while existing approaches often rely on complex and redundant frequency extraction modules or multi-branch parallel frequency extraction architectures that struggle to balance efficiency and representational capacity, it becomes necessary

to introduce a lightweight frequency modeling mechanism that can seamlessly integrate into the Transformer framework. Prompt learning, as a parameter-efficient, lightweight modeling paradigm, offers a viable solution to this challenge. Incorporating a small number of learnable prompt parameters to flexibly guide the model enables efficient integration of prior frequency-domain knowledge into EEG sequences. Building on this foundation, we propose using sparse frequency tokens as frequency-domain prompts for temporal signals. This approach enhances the model's perception and integration capabilities for global frequency components without introducing additional complex structures or disrupting sequence continuity. This design not only effectively reduces computational redundancy and maintains sequence modeling continuity but also helps fully unleash the Transformer's potential in temporal modeling for EEG decoding, thereby providing a novel solution for building more efficient and practical next-generation EEG decoding and classification systems.

The experimental results show that following the introduction of FGPT for frequency prompting, the three baseline models achieve significant outperformance in decoding performance over the original model in cognitive attention [37], fatigue driving [38], and mental cognitive work [39]. Across the three tasks, FGPT elevated the average accuracy of EEG-ViT [15], EEG-Conformer [14], and EEG-Deformer [15] by 1.74%, 1.56%, and 3.87%, respectively. The F1-macro scores were similarly enhanced, with average gains of 1.51%, 1.59%, and 3.90%. Our main contributions are as follows.

- We utilize sparse token prompt learning based on gated fusion to model the frequency components of Transformers while avoiding disrupting the spatio-temporal continuity of the original sequence.
- The core operation of the proposed FGPT model is straightforward, achieving performance improvements without requiring complex structures or additional constraints.
- Our method can be extended to a Transformer-based EEG decoding framework, achieving robust EEG decoding with only a small number of prompt tokens.

## II. METHODOLOGY

In this work, we propose a novel frequency cue tuning method for EEG decoding, suitable for efficient and general-purpose Transformer-based EEG systems. Fig. 1 illustrates the entire frequency prompt learning method, which achieves global frequency component modeling while preserving the original sequence's spatio-temporal integrity using only a small number of sparse frequency tokens. By cleverly integrating spatio-temporal and frequency-domain information, the method leverages frequency-domain information from latent embeddings to expand the search space, thereby enabling the capture of frequency components within the spatio-temporal search space.

### A. Frequency-Gated Prompted Transformer (FGPT)

Prompt learning is commonly used in the field of Natural Language Processing (NLP), and its core idea is to guide a pre-trained model to understand the task and output the desired

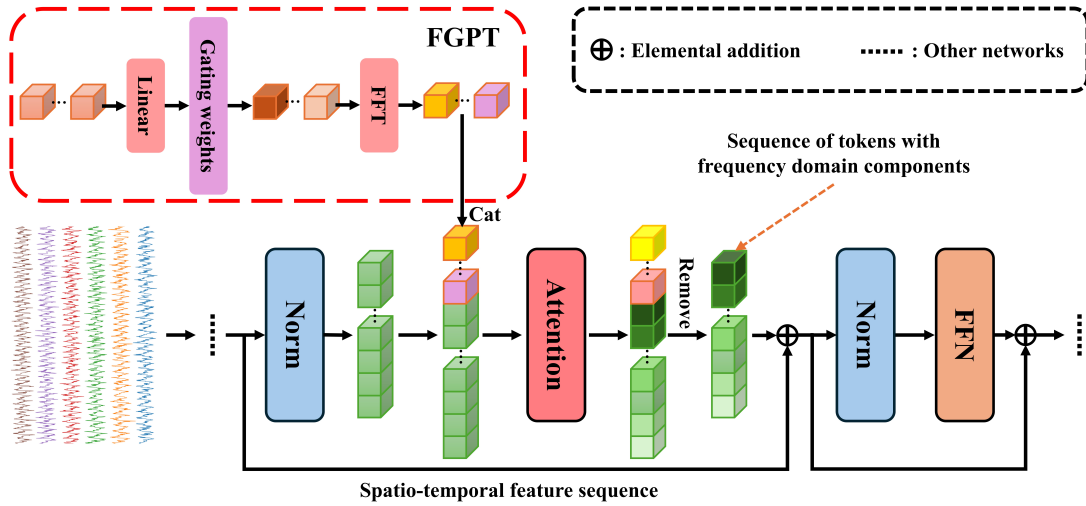


Fig. 1. Sparse frequency tokens are used for prompt fine-tuning of the Transformer network for EEG decoding, enabling global frequency component modeling while preserving the temporal-spatial integrity of the original sequence. The ellipsis symbol represents other network components.

result by designing specific input prompts. We borrow this idea from the NLP domain to improve the robustness of EEG decoding. The use of cue learning in the Transformer often places cue tokens in the Transformer encoding layer, allowing the entire cue engineering to be carried out throughout the Transformer layer. These learnable cue tokens per layer are usually defined as  $P = \{P_1, P_2, \dots, P_n\}$ . Equation (1) shows the sequence of the Transformer after embedding cue tokens per layer.

$$S_1 = L_1^T(P_1, E), \dots, S_i = L_i^T(P_i, S_{i-1}) \quad (1)$$

where  $E$  is the sequence obtained after the embedding layer and  $S_i$  represents the sequence after cue learning  $L_i^T$ .

When processing EEG signals, the models established are typically not pre-trained. Most models first employ convolutional neural networks to perform local feature extraction before feeding the features into the Transformer. In this process, some models retain the embedding layer, while others may discard it, but this does not conflict with the embedding position of prompt tokens. If prompt tokens are embedded following the classical embedding approach, they would be passed through normalization layers, attention layers, and feed-forward neural networks for computation, which increases the computational overhead. In contrast, our method embeds the prompts before the attention calculation, avoiding interference from other components of the Transformer with the original sequence distribution.

Our FGPT model performs sparse frequency prompt token embedding before executing attention calculations. Before embedding, frequency prompt tokens first use a gating mechanism to fuse the weights of the original prompt tokens in order to measure the contributions of different prompt tokens. This calculation process is shown in Equation (2).

$$p_n^i = \text{Linear}(p_n^i) \odot \text{Sigmoid}(w_n^i) \quad (2)$$

A single sparse prompt token  $p_n^i$  undergoes weighted fusion after projection embedding and Sigmoid gating, where  $w_n^i$  is

the learnable weight vector for a single prompt token in the  $i$ -th layer of the Transformer. Subsequently, the sparse prompt token is converted into a frequency-domain prompt token using the Fast Fourier Transform (FFT). FFT, as a powerful algorithm for computing the Discrete Fourier Transform (DFT), can convert finite sequence samples into sequences of equal length in the discrete Fourier transform. DFT can be expressed as Equation (3).

$$\mathcal{F}(x) := X(k) = \sum_{n=0}^{N-1} x_n e^{-j \frac{2\pi kn}{N}}, 0 \leq k \leq N-1 \quad (3)$$

where  $x_n$  represents the given sequence, and  $n$  denotes elements in the interval  $[0, N-1]$ . This transformation calculation process is defined as  $\mathcal{F}$ . The initial DFT has a quadratic computational complexity, which can be reduced to a linear logarithmic level after being decomposed by the FFT butterfly operation. Therefore, we use the FFT to implement the frequency-domain conversion to avoid incurring greater computational overhead.

Sparse prompt tokens are generated using FFT to produce frequency-domain sparse prompt tokens, which are then concatenated with the normalized EEG token sequence, as shown in Equation (4).

$$S_{\mathcal{F}}^i = [\mathcal{R}(\mathcal{F}_P([p_1^i, p_2^i, \dots, p_n^i])), \text{Norm}(S^{i-1})] \quad (4)$$

where  $\mathcal{F}_P$  represents the FFT transformation of the sparse prompt tokens,  $\text{Norm}$  represents the normalization layer,  $S^{i-1}$  represents the EEG token sequence,  $[\dots, \dots]$  represents concatenation, and  $\mathcal{R}$  represents retaining the real part of the FFT output. In EEG analysis, due to the widespread application and clear physiological significance of frequency band energy metrics, while phase information exhibits greater variability across different tasks and individuals, priority is given to energy features to reduce model complexity. Therefore, only the real-part component is retained here. After concatenation, the new token sequence  $S_{\mathcal{F}}^i$  embedding yields the  $Q_{\mathcal{F}}^i$ ,  $K_{\mathcal{F}}^i$ , and

$V_{\mathcal{F}}^i$ , followed by attention calculation with sparse frequency-domain token prompts, thereby achieving global frequency-domain component modelling without altering the continuity of the original sequence. Attention calculation with frequency prompts is as follows.

$$\text{Attn}_{\mathcal{F}}^i = \text{Softmax}\left(\frac{Q_{\mathcal{F}}^i K_{\mathcal{F}}^{i^T}}{\sqrt{d}}\right) V_{\mathcal{F}}^i \quad (5)$$

where  $\sqrt{d}$  is the scaling factor. After the attention calculation is completed, the sparse frequency-domain tokens are removed, and the resulting sequence can be represented as  $S^i$ . The purpose is to prevent the sparse frequency-domain tokens from participating in subsequent normalization layers and feedforward neural network calculations, thereby improving model efficiency.

### B. The Effect of FGPT on Layer Normalization

Through systematic theoretical analysis, we compared the effectiveness of different embedding positions for frequency prompt tokens, including placement before the entire Transformer block, before each Transformer layer, and before each attention layer. We demonstrated the superiority of the strategy that embeds tokens before the attention layer and subsequently removes them. For the first LayerNorm in each Transformer layer, FGPT does not perform any operation. The other two embedding methods modify the number of normalized token samples in LayerNorm, where the sequence can be represented as  $X \in \mathbb{R}^{(T+P) \times D}$ . Equations (6) and (7) provide the mean and standard deviation statistics for a single feature during the computation of the first LayerNorm in the Transformer.

$$\mu_l = \frac{1}{D} \sum_{d=1}^D x_{l,d} \quad (6)$$

$$\sigma_l^2 = \frac{1}{D} \sum_{d=1}^D (x_{l,d} - \mu_l)^2 \quad (7)$$

After calculating the mean and variance, normalize each feature value according to Equation (8).

$$\hat{x}_{l,d} = \frac{x_{l,d} - \mu_l}{\sqrt{\sigma_l^2 + \epsilon}} \quad (8)$$

where  $\epsilon > 0$  is a small constant added for numerical stability. After applying Equation (9), the learnable parameters  $\gamma_d$  and  $\beta_d$  are used to perform an affine transformation to achieve scaling and translation of the normalized features.

$$y_{l,d} = \gamma_d \cdot \hat{x}_{l,d} + \beta_d \quad (9)$$

Since each token is normalized separately, the frequency prompt token does not affect the original token normalization during forward propagation. However, it does affect the learning of feature-level parameters  $\gamma_d$  and  $\beta_d$  in the radiative transformation during backward propagation. This is because the gradient calculation of  $\gamma_d$  and  $\beta_d$  depends on all tokens, and the gradient of the loss function is propagated backward

through all tokens to  $\gamma_d$  and  $\beta_d$ . The backward propagation process at this point is shown in Equation (10) and (11).

$$\frac{\partial \mathcal{L}_{\text{CE}}}{\partial \gamma} = \sum_{l=1}^{L=T+P} \left( \frac{\partial \mathcal{L}_{\text{CE}}}{\partial Y} \cdot \hat{X} \right) \quad (10)$$

$$\frac{\partial \mathcal{L}_{\text{CE}}}{\partial \beta} = \sum_{l=1}^{L=T+P} \frac{\partial \mathcal{L}_{\text{CE}}}{\partial Y} \quad (11)$$

Among them,  $\gamma$  and  $\beta$  represent the corresponding token component representations,  $\hat{X}$  represents the normalized token component representations,  $Y \in \mathbb{R}^{(T+P) \times D}$  represents the final output of LayerNorm, and  $\mathcal{L}_{\text{CE}}$  represents the cross-entropy loss function in the model. When the frequency prompt token is added, the sequence length increases, indicating that the prompt frequency token contributes an additional gradient term. Therefore, the update directions of  $\gamma$  and  $\beta$  are jointly influenced by all tokens, and the introduction of the prompt token causes a shift in the original sequence distribution.

After passing through the first LayerNorm, these features are transmitted to the attention mechanism, where the attention calculation causes the original sequence to contain frequency prompt information. Therefore, when undergoing normalization by the second LayerNorm, the original sequence, which includes frequency-domain prompt information, will exhibit statistical bias when calculating statistical components as in Equations (6) and (7). This will also cause changes in the corresponding learnable parameters. Theoretically, when embedding frequency prompt tokens before feeding them into the Transformer layer, the second LayerNorm has a greater impact on the results compared to the first LayerNorm.

To reduce the statistical bias caused by frequency prompts on the LayerNorm computation process, the FGPT scheme chooses to embed frequency prompts before each layer of attention and remove these frequency prompts after the attention computation is completed. This also avoids the computation of frequency tokens required by feedforward neural networks, thereby reducing the computational overhead of the model.

## III. EXPERIMENT

### A. EEG Datasets and Pre-processing

We evaluated the performance of EEG-FGPT using four commonly used public benchmark datasets. Dataset A [37] pertains to cognitive attention, dataset B [38] to fatigued driving, dataset C [39] to mental cognitive tasks, and dataset D [40] to motor imagery.

1) **Dataset A:** Dataset A provides EEG data recorded from 26 healthy participants performing a Discrimination/Selection Response (DSR) task to assess cognitive attention. Each participant in this dataset undergoes three experimental phases, each containing multiple 40-second attention cycle tasks. A 20-second rest period follows each trial, with tasks and rest periods alternating throughout. The sampling rate of this dataset is 1000 Hz, with a total of 28 EEG channels recorded. EEG activity data were collected using the BrainAmp EEG amplifier (Brain Products GmbH, Gilching, Germany).



2) **Dataset B:** Dataset B utilizes dynamic brain data analyzed from a 90-minute simulated driving task in a virtual reality (VR) driving scenario to assess the cognitive fatigue state of the participating drivers. Dataset B included 27 driver participants, with 32-channel EEG recordings collected while the drivers traveled on a four-lane highway. During driving, participants were instructed to maintain the vehicle at a constant speed in the center of the lane, with random lane deviation events inducing the vehicle to drift left or right. The entire experimental process included events such as deviation onset, reaction onset, and reaction end. In the next experiment, participants were required to drive back to the original cruising lane, with this experiment beginning 5–10 seconds after the previous experiment concluded.

3) **Dataset C:** Dataset C provides 19-channel EEG activity from 36 subjects performing a mental cognitive load task, showing subjects in a state of high-intensity cognitive activity while performing continuous subtraction. During EEG recording, subjects sat in a dark, soundproof room and leaned comfortably in an armchair. Prior to the experiment, participants were instructed to relax as much as possible and were informed about the arithmetic task, which required them to perform mental calculations quickly without speaking. After adapting to the experimental conditions for 3 minutes, participants performed a 4-minute continuous subtraction mental arithmetic task.

4) **Dataset D:** Dataset D records electroencephalogram signals from nine subjects performing four motor imagery tasks (left hand, right hand, both feet, tongue). Data acquisition utilized 22 Ag/AgCl electrodes arranged according to the International 10-20 System, sampled at 250 Hz, and preprocessed with bandpass filtering between 0.5 and 100 Hz. The entire dataset comprises two experimental sessions: the first session serves as the training set, and the second session as the test set. Each session contains 6 runs, with each run including 48 trials. The number of trials across the four task categories is evenly distributed.

5) **Pre-processing:** Given FGPT's simplicity and scalability, we did not design independent preprocessing for most datasets, applying specialized processing only to dataset D to accommodate different models. For datasets A, B, and C, we retained the preprocessing from the original model [15] for subsequent comparisons. For dataset D, we used data from the first session, excluded artifact trials, retained both left-hand and right-hand motor imagery tasks, set the time window to 0–4 seconds, applied bandpass filtering from 0.5 to 100 Hz, maintained a sampling rate of 250 Hz, and used 22 EEG channels as input.

## B. Transformer-based and Frequency-based Baselines

We used three EEG decoding models based on the Transformer framework and demonstrated their effectiveness by comparing the results before and after adding our FGPT, as well as those based on the Transformer framework. The three models are EEG-ViT [15], EEG-Conformer [14], and EEG-Deformer [15]. The EEG-ViT model is an adapted version of Vision Transformer (ViT) [41]. Additionally, we

evaluated two classic frequency feature extraction methods specifically designed for motor imagery (MI) tasks as representative baselines of distinct topological structures. These include FACT-Net [42], which incorporates a frequency adapter, and ADFCNN [43], which extracts multi-frequency features through a dual-branch architecture. Although various other highly competitive frequency feature extraction methods exist, these two were selected as baseline representatives to reflect fundamentally different topological designs.

1) **EEG-ViT:** EEG-ViT is a model that processes EEG signals using a purely ViT-based architecture [41]. Specifically, EEG signals are divided into non-overlapping segments along the time dimension, and then these non-overlapping segments are converted into tokens using linear mapping. This processing allows ViT to effectively process EEG data while retaining the temporal continuity of the token sequence.

2) **EEG-Conformer:** EEG-Conformer is an EEG decoding model based on the Transformer framework. It specifically uses CNN to capture local features and Transformer to model long-range dependencies in EEG signals. This combination of CNN and Transformer can extract spatio-temporal features from EEG signals. Specifically, it uses a one-dimensional CNN to extract fine-grained features and self-attention to reveal global correlations in EEG time series.

3) **EEG-Deformer:** EEG-Deformer was developed to address the issue that most models cannot capture the dynamic evolution of EEG signal features from coarse to fine-grained levels. It innovatively introduces two components into the CNN-Transformer framework. The first is the Hierarchical Coarse-to-Fine Transformer (HCT) module, which integrates a Fine-Grained Time Learning (FTL) branch into the Transformer, effectively identifying coarse-to-fine temporal patterns. The other is a dense information purification module, which performs multi-layer purification of temporal information to achieve high-precision EEG decoding.

4) **FACT-Net:** FACT-Net addresses the issue of inadequate feature extraction in existing models when processing non-stationary EEG signals, where fixed parameters struggle to capture dynamic frequency-domain variations. It innovatively introduces two core components within the CNN architecture. One is the Frequency Adapter, a module that employs a learnable filtering mechanism to achieve adaptive weighting and reconstruction of features across different frequency bands, precisely pinpointing key frequency-domain information during motor imagery. The other is a time-domain periodic Inception module, which utilizes multi-scale branches to uncover deep periodic patterns within EEG signals while maintaining lightweight inference, thereby achieving efficient EEG decoding.

5) **ADFCNN:** ADFCNN addresses the limitation of traditional models in processing EEG signals, where a fixed receptive field struggles to simultaneously capture high-frequency details and low-frequency contour features. It innovatively introduces two core components within its network architecture. One is the dual-scale branch frequency extraction module, which captures both fine-grained high-frequency fluctuations and macro-level low-frequency trends in EEG signals by deploying convolutional kernels at parallel scales, enabling mul-

tidimensional spectral information representation. The other is an attention-based feature fusion mechanism. This module adaptively weights the heterogeneous features extracted by the dual branches, dynamically selecting the most discriminative components while suppressing background noise. This enhances decoding accuracy in complex motor imagery tasks.

### C. Experiment Settings

To ensure fair comparison, we strictly followed the original experimental configurations of each comparison model in conducting all experiments. The experiments adhered to the general principle of thematic independence to ensure that no test data information was exposed during the training phase. A leave-one-subject-out (LOSO) cross-validation strategy was implemented for the four EEG datasets (A, B, C, and D). In each iteration, the EEG data from a single subject was used as the test set, while the remaining data was divided into a training set and a validation set in an 80% to 20% ratio. The optimal model from the validation set was ultimately selected for testing. Model performance was evaluated using two metrics: accuracy (ACC) and macro-F1 score (Macro-F1). The specific calculation process for ACC is detailed in Equation (12).

$$ACC = \frac{TP + TN}{TP + FP + TN + FN} \quad (12)$$

The specific calculation process for Macro-F1 is detailed in Equation (13).

$$\begin{aligned} F1-macro &= \frac{1}{M} \sum_{m=1}^M F1-score_m \\ &= \frac{1}{M} \sum_{m=1}^M \frac{2 \cdot TP_m}{2 \cdot TP_m + FP_m + FN_m} \end{aligned} \quad (13)$$

In these Equations,  $TP$  represents true positives,  $FP$  represents false positives,  $FN$  represents false negatives, and  $m$  represents different classes. For the choice of loss function, we adopt cross-entropy as the loss function. Since the cross-entropy loss function directly optimizes the difference between the predicted distribution and the true distribution through probabilistic modeling, its characteristic of gradient magnitude being proportional to error significantly accelerates model convergence. At the same time, with the framework's highly optimized single-step computation and natural support for multi-class classification extension, it has become the most effective and widely used loss function in classification tasks. Its calculation process is shown in Equation (14).

$$\mathcal{L}_{CE} = -\frac{1}{N} \sum_{n=1}^N \log \left( \frac{\exp(x_{n,y_n})}{\sum_{c=1}^C \exp(x_{n,c})} \right) \quad (14)$$

where  $N$  is the number of samples,  $C$  is the number of categories,  $x_n$  is the predicted value, and  $y_n$  is the index of the true value.

We completed all experiments, including model training and testing, on a single NVIDIA RTX 2080 Ti GPU. The experiments used the Adam optimizer with an initial learning rate of  $1e-3$ , and the learning rate was dynamically adjusted using

a cosine annealing strategy. The batch size for all datasets input to the model was uniformly set to 64. The model was trained for 200 epochs, and finally, the model weights with the highest accuracy on the validation set were selected for testing. To ensure the reproducibility of the experiments, the random seed for all experiments was fixed at 2023. Furthermore, for the proposed FGPT module, its internal learnable prompt embeddings and gating parameters were trained synchronously using the same optimizer and learning rate as the main network, without introducing additional hyperparameters. In the gating fusion mechanism, the temperature coefficient of the Sigmoid activation function was fixed at 1.0. This concise design ensures the easy transferability and training stability of FGPT across different Transformer architectures.

## IV. RESULTS AND ANALYSIS

### A. Experimental Results

We added FGPT to three Transformer-based methods and validated them on three EEG datasets. TABLE I compares the performance results before and after using FGPT, including the specific number of sparse prompt tokens. The results demonstrate that our proposed FGPT strategy delivers outstanding performance. After applying FGPT, both accuracy and F1-macro improved, while the standard deviation decreased in the vast majority of results. Particularly on Dataset A, EEG-Deformer achieved a 6.53% increase in ACC and a 6.46% improvement in F1-macro after FGPT implementation. It is worth noting that, as shown in TABLE IV, using just one frequency prompt on Dataset A resulted in a 5.45% increase in accuracy and a 5.12% improvement in the F1-macro metric compared to the original EEG-Deformer model in TABLE I. Across these nine experiments, seven achieved ACC gains exceeding 1.50%, five surpassed 2.00%, and three exceeded 3.00%. For F1-macro, eight out of nine experiments showed improvements exceeding 1.00%, with four achieving gains over 2.00% and the highest improvement surpassing 6.00%. These results demonstrate FGPT's robustness, attributed to its sparse frequency prompt tokens that enable frequency component modeling without disrupting the continuity of the original sequence. Our results demonstrate that FGPT enhances the decoding capabilities of various models, including the foundational EEG-ViT and the more advanced EEG-Conformer and EEG-Deformer. A positive correlation is observed between the performance gain and the architectural complexity of these models. This correlation is likely attributable to the superior feature extraction capabilities of advanced models, which provide FGPT with more robust representations of subject intent. It follows that the integration of FGPT into state-of-the-art architectures holds considerable promise for achieving peak decoding performance.

Regarding the number of frequency prompt tokens, the quantity of prompt tokens does not exceed five. Based on the experimental results, we speculate that there may be a certain relationship between the number of tokens and specific frequency bands in EEG, indirectly suggesting that the frequency-prompted tokens corresponding to these bands play a positive role in the neural decoding representation of EEG information.

TABLE I

A COMPARISON OF THE PERFORMANCE OF THREE TRANSFORMER-BASED METHODS ON THREE EEG DATASETS BEFORE AND AFTER USING FGPT, AS WELL AS THE NUMBER OF PROMPT TOKENS USED WITH FGPT.

| Method        |                    | Dataset A: Attention |       |          |       | Dataset B: Fatigue |       |          |       | Dataset C: Mental Workload |       |          |       |
|---------------|--------------------|----------------------|-------|----------|-------|--------------------|-------|----------|-------|----------------------------|-------|----------|-------|
|               |                    | ACC                  | std   | F1-macro | std   | ACC                | std   | F1-macro | std   | ACC                        | std   | F1-macro | std   |
| EEG-ViT       | EEG-ViT            | 67.39                | 7.48  | 66.29    | 9.13  | 66.81              | 13.85 | 64.22    | 13.44 | 63.94                      | 15.02 | 61.67    | 16.36 |
|               | EEG-ViT+FGPT       | 67.81                | 6.96  | 66.80    | 8.10  | 69.84              | 12.21 | 66.93    | 11.69 | 65.71                      | 14.12 | 62.99    | 16.44 |
|               | Improvement        | 0.42↑                | 0.52↓ | 0.51↑    | 1.03↓ | 3.03↑              | 1.64↓ | 2.71↑    | 1.75↓ | 1.77↑                      | 0.90↓ | 1.32↑    | 0.08↑ |
|               | Token number       | 4                    |       |          |       | 3                  |       |          |       | 5                          |       |          |       |
| EEG-Conformer | EEG-Conformer      | 79.36                | 7.98  | 78.81    | 8.78  | 74.93              | 11.16 | 72.17    | 12.71 | 68.68                      | 15.13 | 64.99    | 18.76 |
|               | EEG-Conformer+FGPT | 80.24                | 6.76  | 79.86    | 7.27  | 76.49              | 10.11 | 73.44    | 12.51 | 70.93                      | 14.22 | 67.43    | 18.46 |
|               | Improvement        | 0.88↑                | 1.22↓ | 1.05↑    | 1.51↓ | 1.56↑              | 1.05↓ | 1.27↑    | 0.20↓ | 2.25↑                      | 0.91↓ | 2.44↑    | 0.30↑ |
|               | Token number       | 1                    |       |          |       | 5                  |       |          |       | 4                          |       |          |       |
| EEG-Deformer  | EEG-Deformer       | 77.22                | 8.24  | 76.66    | 8.70  | 75.41              | 10.88 | 72.97    | 12.51 | 71.41                      | 15.28 | 68.30    | 18.95 |
|               | EEG-Deformer+FGPT  | 83.75                | 8.62  | 83.12    | 9.97  | 77.47              | 8.29  | 74.31    | 11.26 | 74.43                      | 14.74 | 72.20    | 17.66 |
|               | Improvement        | 6.53↑                | 0.38↑ | 6.46↑    | 1.27↑ | 2.06↑              | 2.59↓ | 1.34↑    | 1.25↓ | 3.02↑                      | 0.54↓ | 3.90↑    | 1.29↓ |
|               | Token number       | 5                    |       |          |       | 1                  |       |          |       | 5                          |       |          |       |

TABLE II

RESULTS OF ABLATION STUDIES CONDUCTED ON THREE BENCHMARK DATASETS. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD.

| Method        |                 | Dataset A: Attention |      |              |      | Dataset B: Fatigue |       |              |       | Dataset C: Mental Workload |       |              |       |
|---------------|-----------------|----------------------|------|--------------|------|--------------------|-------|--------------|-------|----------------------------|-------|--------------|-------|
|               |                 | ACC                  | std  | F1-macro     | std  | ACC                | std   | F1-macro     | std   | ACC                        | std   | F1-macro     | std   |
| EEG-ViT       | w/o Gate Fusion | 67.15                | 6.92 | 66.09        | 8.21 | 67.10              | 13.89 | 64.34        | 13.13 | 63.98                      | 13.17 | 61.43        | 14.93 |
|               | w/o FFT Prompts | 67.18                | 6.24 | 65.92        | 7.74 | 67.60              | 14.05 | 65.00        | 13.74 | 63.03                      | 14.81 | 60.75        | 16.33 |
|               | FGPT (Ours)     | <b>67.81</b>         | 6.96 | <b>66.80</b> | 8.10 | <b>69.84</b>       | 12.21 | <b>66.93</b> | 11.69 | <b>65.71</b>               | 14.12 | <b>62.99</b> | 16.44 |
|               | Token number    | 4                    |      |              |      | 3                  |       |              |       | 5                          |       |              |       |
| EEG-Conformer | w/o Gate Fusion | 79.14                | 8.20 | 78.27        | 9.95 | 75.13              | 10.27 | 72.45        | 11.48 | 69.78                      | 14.53 | 65.90        | 18.88 |
|               | w/o FFT Prompts | 79.25                | 7.76 | 78.65        | 8.66 | 75.54              | 9.09  | 72.52        | 10.73 | 69.25                      | 15.24 | 65.49        | 19.18 |
|               | FGPT (Ours)     | <b>80.24</b>         | 6.76 | <b>79.86</b> | 7.27 | <b>76.49</b>       | 10.11 | <b>73.44</b> | 12.51 | <b>70.93</b>               | 14.22 | <b>67.43</b> | 18.46 |
|               | Token number    | 1                    |      |              |      | 5                  |       |              |       | 4                          |       |              |       |

Additionally, the sparse number of frequency prompt tokens reduces the computational burden on the model, facilitating simpler model design.

## B. FGPT Ablation Study

1) *Effective gate fusion*: We conducted ablation studies on EEG-FGPT using two models, EEG-ViT and EEG-Conformer, across three EEG datasets. In TABLE II, the removal of the gated fusion component from EEG-FGPT in both EEG-ViT and EEG-Conformer resulted in varying degrees of impact on the final ACC and F1-macro scores. For EEG-ViT, the most significant impact was observed on Dataset B, where ACC decreased by 2.74% and F1-macro dropped by 2.59%. The smallest effect occurred on Dataset A, with ACC reduced by 0.66% and F1-macro by 0.71%. In EEG-Conformer, the largest ACC reduction was 1.36% on Dataset B, while the smallest was 1.10% on Dataset A. Regarding F1-macro, the greatest decrease was 1.59% on Dataset A, followed by 0.99% on Dataset B. These experimental results demonstrate the crucial role of gated fusion in EEG-FGPT, which dynamically and adaptively weights the contributions of different frequency prompt tokens and establishes a stable and adaptive gating gradient flow.

2) *Effective frequency prompts*: Similarly, we conducted ablation experiments on frequency prompts to validate their effectiveness. TABLE II presents the ablation results for two Transformer-based models across three datasets. In EEG-ViT's ACC, frequency prompts had the greatest impact on dataset C, achieving 2.68%, while dataset A saw the smallest effect, decreasing by 0.63%. In EEG-ViT's F1-macro, frequency prompts had the greatest impact on dataset C, reaching 2.24%, and the smallest impact on dataset A, decreasing by 0.88%. In EEG-Conformer's ACC, frequency prompts had the greatest impact on dataset C, reaching 1.68%, and the smallest impact on dataset B, decreasing by 0.95%. In EEG-Conformer's F1-macro, frequency hints had the greatest impact on dataset C, reaching 1.94%, while affecting dataset B the least, decreasing by 0.92%. The ablation experiment results for frequency hints demonstrate their effectiveness. By highlighting important frequencies in the original spatio-temporal sequence, they enable frequency component modeling within the spatio-temporal framework, thereby enhancing the robustness of EEG decoding performance.

3) *Effective frequency prompt embedding positions*: Different embedding positions for sparse frequency tokens may lead to different results, so we designed an ablation study for the embedding positions of prompt tokens, including Pre-

TABLE III

PERFORMANCE COMPARISON OF SPARSE FREQUENCY TOKENS AT DIFFERENT EMBEDDING POSITIONS. THE BEST RESULTS ARE MARKED IN BOLD.

| Method  |                              | Dataset A: Attention |      |              |      | Dataset B: Fatigue |       |              |       | Dataset C: Mental Workload |       |              |       |
|---------|------------------------------|----------------------|------|--------------|------|--------------------|-------|--------------|-------|----------------------------|-------|--------------|-------|
|         |                              | ACC                  | std  | F1-macro     | std  | ACC                | std   | F1-macro     | std   | ACC                        | std   | F1-macro     | std   |
| EEG-ViT | Pre-Transformer (All Layers) | 67.60                | 7.06 | 66.46        | 8.87 | 66.38              | 13.61 | 63.47        | 12.84 | 65.37                      | 14.18 | <b>63.01</b> | 16.17 |
|         | Pre-Transformer Block        | 67.66                | 6.66 | 66.67        | 7.90 | 67.11              | 15.47 | 64.51        | 15.05 | 63.84                      | 13.01 | 61.16        | 15.17 |
|         | Pre-Attention (All Layers)   | <b>67.81</b>         | 6.96 | <b>66.80</b> | 8.10 | <b>69.84</b>       | 12.21 | <b>66.93</b> | 11.69 | <b>65.71</b>               | 14.12 | 62.99        | 16.44 |
|         | Token number                 | 4                    |      |              |      | 3                  |       |              |       | 5                          |       |              |       |

TABLE IV

PERFORMANCE COMPARISON OF SPARSE FREQUENCY TOKENS WITH DIFFERENT EMBEDDING NUMBERS. THE BEST RESULTS ARE MARKED IN BOLD.

| Method        |                 | Dataset A: Attention |       |              |       | Dataset B: Fatigue |       |              |       | Dataset C: Mental Workload |       |              |       |
|---------------|-----------------|----------------------|-------|--------------|-------|--------------------|-------|--------------|-------|----------------------------|-------|--------------|-------|
|               |                 | ACC                  | std   | F1-macro     | std   | ACC                | std   | F1-macro     | std   | ACC                        | std   | F1-macro     | std   |
| EEG-ViT       | Token number: 1 | 67.47                | 7.24  | 66.04        | 9.12  | 67.69              | 13.81 | 63.89        | 13.38 | 62.98                      | 13.80 | 60.01        | 16.38 |
|               | Token number: 2 | 67.17                | 6.90  | 66.07        | 8.52  | 66.65              | 14.61 | 63.97        | 14.37 | 64.80                      | 14.47 | 62.94        | 15.91 |
|               | Token number: 3 | 67.39                | 6.70  | 66.22        | 8.27  | <b>69.84</b>       | 12.21 | <b>66.93</b> | 11.69 | 64.99                      | 16.10 | 62.77        | 18.00 |
|               | Token number: 4 | <b>67.81</b>         | 6.96  | <b>66.80</b> | 8.10  | 67.47              | 13.75 | 65.34        | 13.19 | 61.59                      | 14.83 | 58.96        | 16.48 |
|               | Token number: 5 | 66.19                | 7.08  | 64.94        | 8.34  | 69.15              | 11.31 | 66.31        | 11.17 | <b>65.71</b>               | 14.12 | <b>62.99</b> | 16.44 |
| EEG-Conformer | Token number: 1 | <b>80.24</b>         | 6.76  | <b>79.86</b> | 7.27  | 74.75              | 10.86 | 71.77        | 12.45 | 69.06                      | 14.34 | 65.22        | 18.74 |
|               | Token number: 2 | 79.88                | 7.26  | 79.47        | 7.85  | 73.91              | 10.18 | 71.04        | 11.90 | 69.11                      | 13.32 | 66.01        | 16.69 |
|               | Token number: 3 | 79.43                | 7.66  | 78.75        | 8.87  | 75.24              | 8.67  | 71.49        | 11.30 | 69.40                      | 14.24 | 65.31        | 18.86 |
|               | Token number: 4 | 78.63                | 8.85  | 77.58        | 10.95 | 74.55              | 10.15 | 71.26        | 12.21 | <b>70.93</b>               | 14.22 | <b>67.43</b> | 18.46 |
|               | Token number: 5 | 78.48                | 8.12  | 77.46        | 10.58 | <b>76.49</b>       | 10.11 | <b>73.44</b> | 12.51 | 68.92                      | 15.62 | 65.40        | 19.29 |
| EEG-Deformer  | Token number: 1 | 82.67                | 9.50  | 81.78        | 11.35 | <b>77.46</b>       | 8.29  | 74.31        | 11.26 | 72.65                      | 15.48 | 69.81        | 18.74 |
|               | Token number: 2 | 81.81                | 10.53 | 80.58        | 13.51 | 76.76              | 7.99  | 73.43        | 10.99 | 72.70                      | 16.93 | 69.45        | 20.96 |
|               | Token number: 3 | 82.80                | 10.30 | 81.71        | 12.92 | 77.38              | 8.32  | 74.30        | 10.84 | 73.37                      | 14.74 | 70.62        | 18.40 |
|               | Token number: 4 | 83.16                | 9.59  | 82.28        | 11.63 | 76.97              | 9.49  | <b>74.35</b> | 11.73 | 72.89                      | 16.06 | 69.97        | 19.48 |
|               | Token number: 5 | <b>83.75</b>         | 8.62  | <b>83.12</b> | 9.97  | 77.29              | 8.21  | 74.01        | 10.66 | <b>74.43</b>               | 14.74 | <b>72.20</b> | 17.66 |

Average Test Loss vs Token Number Across Three Datasets

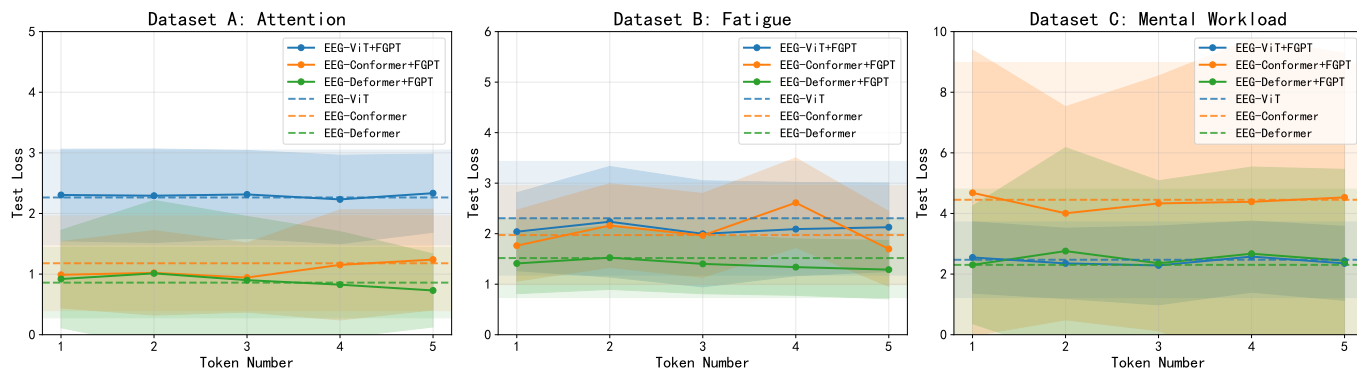


Fig. 2. The relationship between average test loss and prompt token number for three datasets. The line chart depicts loss values, with semi-transparent areas indicating standard deviation ranges.

Transformer (All Layers), Pre-Transformer Block, and Pre-Attention (All Layers). TABLE III compares the performance of different embedding positions for sparse frequency tokens. The results show that different embedding positions lead to different outcomes, with the best results achieved by embedding ACC and F1-macro before all layers of attention calculation, outperforming the other two commonly used em-

bedding position methods. This is because embedding the frequency prompt before attention ensures the continuity of the original sequence and completes the modeling of the frequency component. In contrast, the other two embedding positions interfere with the integrity of the original sequence when passing through the normalization layer and feedforward neural network. Specifically, when passing through the nor-



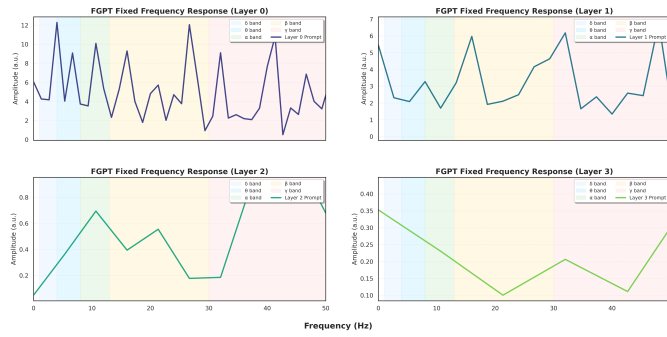


Fig. 3. Subject 44 in Dataset B (Fatigue) exhibits a fixed frequency response in the FGPT within the EEG-Deformer.

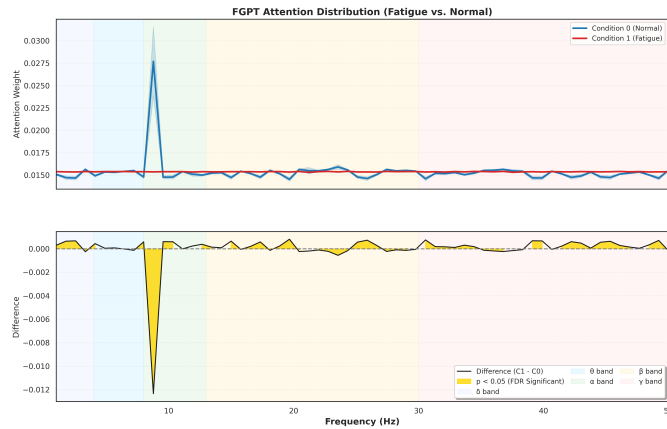


Fig. 4. Dynamic attention distribution of FGPT in EEG-Deformer for subject 44 in dataset B (fatigue).

malization layer, the frequency prompt token is included in the normalization calculation of the entire sequence, which affects the independence of the original sequence.

4) *Effective number of frequency prompt embeddings*: The number of prompt tokens at different frequencies may affect the model's decoding results. TABLE IV compares the performance of EEG-ViT, based on the original Transformer, and EEG-Deformer, based on a CNN-Transformer hybrid structure, under different numbers of prompt tokens at different frequencies. We set up experiments with 1 to 5 prompt tokens and found that while increasing the number of tokens may improve performance, it also significantly increases the computational overhead of attention calculations. Experimental results show that in both methods, using only a small number of prompt tokens can achieve robust decoding performance across three datasets. We found that the number of prompt tokens has a significant impact on decoding performance, which is closely related to the primary frequency components in the original sequence effectively distinguishing between different subjects and classification tasks. This result validates the effectiveness of the sparse token prompting method, which achieves high-performance EEG decoding at low computational cost by introducing only a small number of frequency prompt tokens. Additionally, it does not alter the original model structure, demonstrating strong scalability.

To comprehensively compare the performance of prompts with varying token frequencies, Fig. 2 presents the mean

test loss and its standard deviation on the baseline model across three datasets for different prompt token counts. Results indicate that in most cases, introducing FGPT reduces both loss values and standard deviation. Notably, the settings yielding the lowest loss and standard deviation largely align with the optimal accuracy and F1-macro values in TABLE IV, further validating FGPT's superior performance and robustness. Additionally, we observe that the token configuration yielding the smallest loss does not necessarily correspond to the highest classification accuracy. This discrepancy may stem from inter-subject variations in class distributions across datasets, representing a normal phenomenon. Comprehensive analysis indicates that FGPT demonstrates stable and superior performance across most settings, particularly at the optimal prompt token count, where both loss and standard deviation are markedly lower than those of the baseline model.

5) *Effective neural frequency prompts*: To validate the effectiveness of FGPT in neural frequency prompting, Fig. 3 illustrates the evolution of fixed frequency preferences across different layers within the model. As shown, the shallow network (Layer 0) exhibits extremely high response amplitudes, with response peaks precisely concentrated in the  $\alpha$  band around 10 Hz and lower-frequency rhythms. This indicates that the model performs preliminary feature extraction similar to a biological filter on the raw EEG signal during its initial stages. As the network deepens (Layers 1–3), the center of gravity in the response undergoes a nonlinear shift from the low-frequency region toward the high-frequency  $\gamma$  band above 30 Hz, accompanied by a significant compression in response amplitude. This phenomenon confirms FGPT's hierarchical abstraction capability from raw physical frequencies to higher-order global semantics, aligning with the general principle of deep learning's feature extraction from shallow to deep levels. This validates the model's effectiveness in constructing hierarchical representations in the frequency domain.

To further elucidate the role of FGPT in attention guidance, Fig. 4 illustrates the model's attention distribution across different frequency bands following prompt learning. The upper attention map reveals sustained high attention to the  $\alpha$  band (8–13 Hz), which is closely associated with changes in brain rhythms during fatigue states. The lower difference map statistically compares attention distributions between Condition 1 (fatigued state) and Condition 0 (normal state), revealing significant differences in specific frequency bands (highlighted in gold). These primarily manifest as suppression zones in the  $\alpha$  band and compensatory activity regions in higher frequencies. This shift in attention distribution aligns closely with the evolution of neuronal rhythms during fatigued driving, further confirming that the FGPT module not only extracts general frequency features but also adaptively captures key physiological markers directly relevant to fatigue state discrimination.

6) *Effective cross-sport motor imagery dataset prompts*: In the task of motor imagery, which exhibits distinct frequency response characteristics, we validated the superiority of FGPT through the two sets of experiments presented in Table V. Within FACT-Net, replacing the original DFT-IDFT frequency adapter with FGPT (4 tokens) for weight multiplication in-

### Computational Cost (FLOPs) Analysis of EEG Classification Models Across Three Datasets

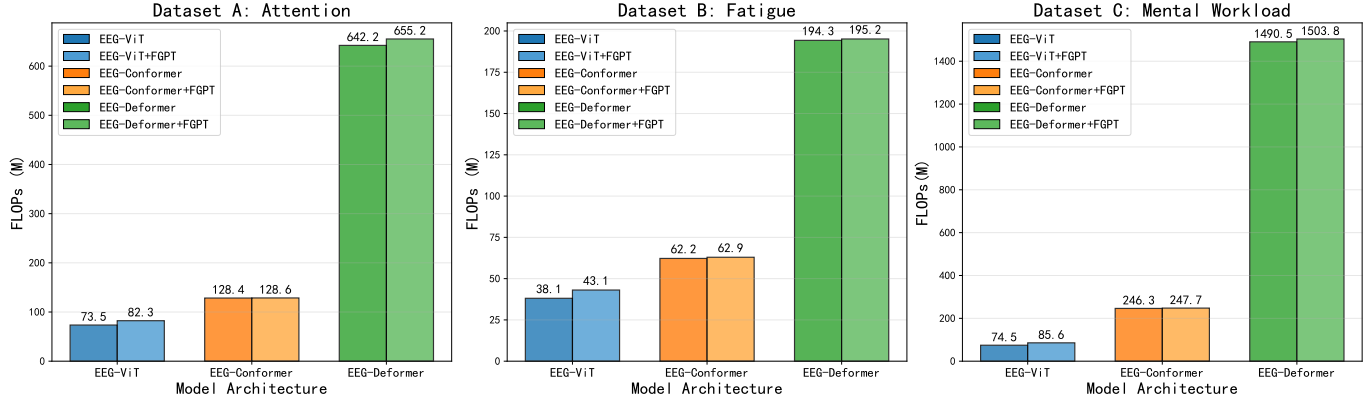


Fig. 5. Comparison and analysis of computational cost (FLOPs) for baseline models before and after FGPT application across three datasets.

### ACC-FLOPs-Params 3D Efficiency Analysis of EEG Classification Models Across Three Datasets

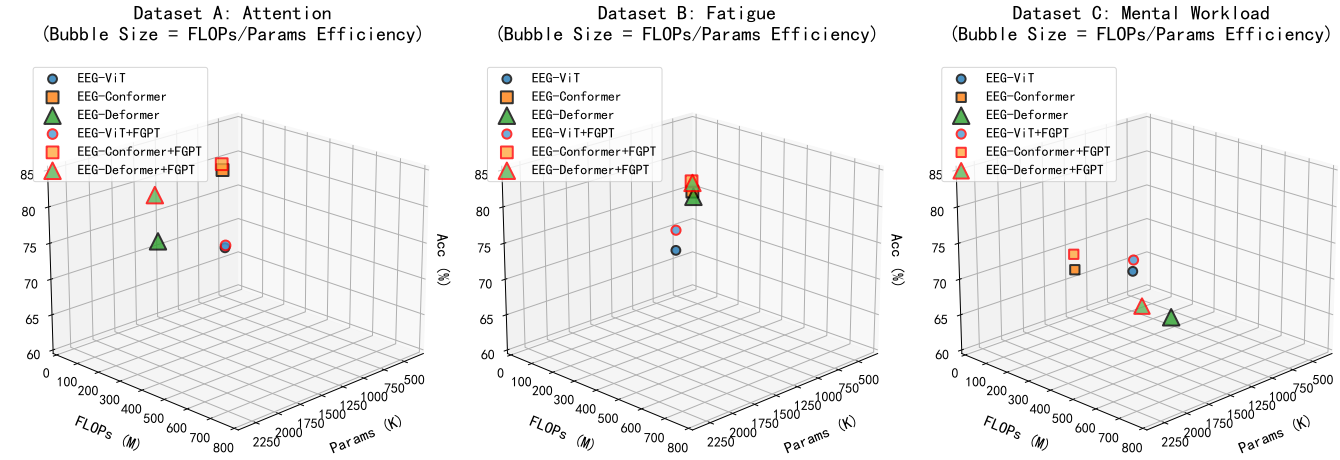


Fig. 6. Comparison and analysis of Accuracy-FLOPs-Params metrics for baseline models across three datasets before and after FGPT application. Models are depicted with distinct colors and markers, while bubble sizes encode the FLOPs/Params ratio, representing overall model efficiency.

TABLE V

THE PERFORMANCE OF DIFFERENT FREQUENCY FEATURE EXTRACTION MODELS WAS COMPARED ON DATASET D (MOTOR IMAGERY), SPECIFICALLY INCLUDING SINGLE-BRANCH AND DUAL-BRANCH MODELS. THE BEST RESULTS ARE MARKED IN BOLD.

| Dataset D: Motor Imagery | FACT-Net |                          | ADFCNN   |                  |
|--------------------------|----------|--------------------------|----------|------------------|
|                          | Original | Repl. w/ FGPT (4 tokens) | Original | +FGPT (4 tokens) |
| ACC                      | 65.66    | <b>66.67</b>             | 64.43    | <b>72.92</b>     |
| std                      | 8.72     | <b>8.66</b>              | 12.56    | <b>8.64</b>      |
| F1-macro                 | 65.08    | <b>66.04</b>             | 57.24    | <b>72.44</b>     |
| std                      | 9.01     | <b>8.79</b>              | 18.94    | <b>8.90</b>      |

creased accuracy from 65.66% to 66.67%, elevated F1-macro from 65.08% to 66.04%, and reduced standard deviation from 8.72 to 8.66. Within the ADFCNN dual-branch frequency feature extraction model, introducing a Transformer with FGPT for joint fine-tuning of frequency features after dual-branch feature merging elevated accuracy from 64.43% to 72.92%,

with F1-macro surging from 57.24% to 72.44%—representing gains of 8.49% and 15.20% respectively. The standard deviations were significantly reduced from 12.56 and 18.94 to 8.64 and 8.90 respectively. Experimental results demonstrate that FGPT, functioning as a frequency prompt token, effectively guides the Transformer's attention mechanism to focus on critical frequency components. Whether replacing traditional frequency adapters or serving as a feature post-processing module, it achieves deep integration and optimised fine-tuning of frequency features, enhancing both model performance and stability.

### C. Performance comparison between the baseline models and those trained with FGPT

Comparative experiments across four datasets demonstrate that FGPT exhibits significant advantages across different backbone models and datasets. In terms of classification accuracy, the introduction of FGPT enhances the performance

TABLE VI  
ANALYSIS OF THE STATISTICAL SIGNIFICANCE OF FGPT PERFORMANCE ON DATASETS A, B, AND C.

| Method        |                    | Dataset A: Attention |            | Dataset B: Fatigue |            | Dataset C: Mental Workload |            |
|---------------|--------------------|----------------------|------------|--------------------|------------|----------------------------|------------|
|               |                    | ACC                  | F1-macro   | ACC                | F1-macro   | ACC                        | F1-macro   |
| EEG-ViT       | Original (Mean±SD) | 68.49±0.85           | 67.56±0.90 | 67.33±0.76         | 64.73±0.90 | 63.57±0.22                 | 61.39±0.37 |
|               | + FGPT (Mean±SD)   | 68.35±0.33           | 67.31±0.43 | 68.27±0.51         | 65.80±0.67 | 64.48±1.06                 | 62.26±1.18 |
| EEG-Conformer | Original (Mean±SD) | 78.88±0.15           | 78.17±0.50 | 74.16±0.26         | 71.32±0.29 | 68.69±1.77                 | 64.79±1.85 |
|               | + FGPT (Mean±SD)   | 79.82±0.20           | 79.39±0.25 | 75.29±2.09         | 72.33±2.01 | 69.05±0.99                 | 65.35±1.27 |
| EEG-Deformer  | Original (Mean±SD) | 79.81±2.45           | 79.27±2.63 | 75.25±1.30         | 72.64±0.85 | 70.40±0.46                 | 67.32±0.54 |
|               | + FGPT (Mean±SD)   | 83.73±0.27           | 83.02±0.32 | 77.47±0.14         | 74.53±0.25 | 72.88±0.92                 | 69.92±1.14 |
| Overall       | Paired t-score     | 5.043036             | 4.234926   | 3.485025           | 3.280551   | 2.884428                   | 2.732828   |
|               | Paired p-value     | 0.000001             | 0.000033   | 0.000738           | 0.001436   | 0.004185                   | 0.006625   |
|               | Significance       | ✓                    | ✓          | ✓                  | ✓          | ✓                          | ✓          |

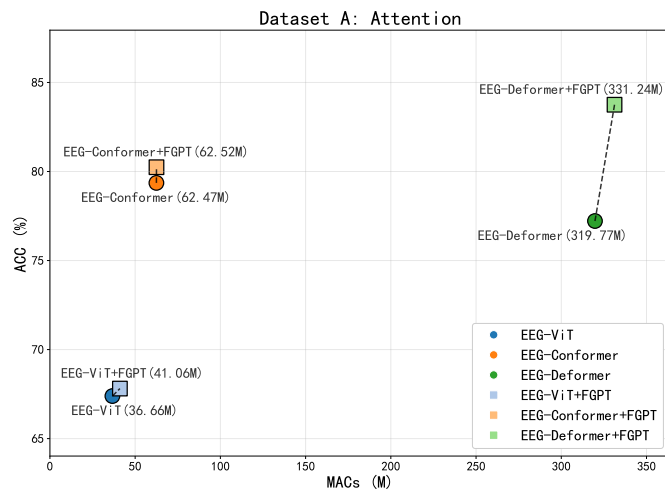


Fig. 7. Accuracy-Computation Complexity scatter plot comparison of three Transformer-Based EEG decoding models on the Attention dataset before and after applying FGPT.

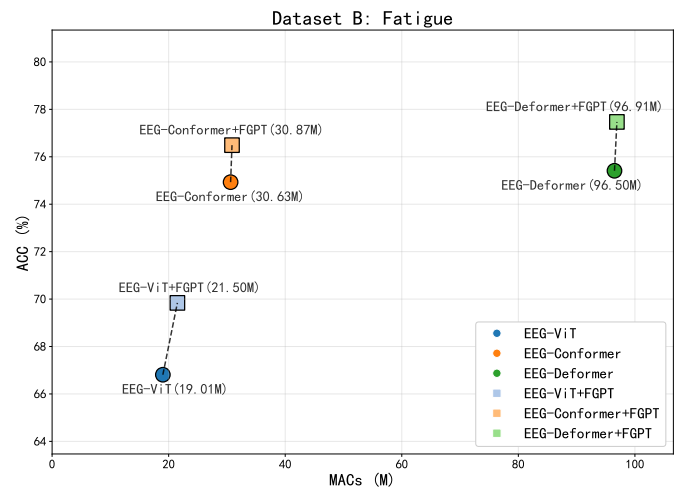


Fig. 8. Accuracy-Computation Complexity scatter plot comparison of three Transformer-Based EEG decoding models on the Fatigue dataset before and after applying FGPT.

of all models. Taking EEG-Deformer as an example, accuracy on Dataset A increased from 77.22% to 83.75%, representing a 6.53% improvement. On Datasets B and C, EEG-ViT combined with FGPT achieved gains of 3.03% and 1.77% respectively, demonstrating FGPT's effectiveness even for smaller models. Crucially, these performance improvements entail only minimal computational overhead. As illustrated in Fig. 5, EEG-Conformer on Dataset A saw a mere 0.15% increase in FLOPs while achieving a 0.88% accuracy gain. EEG-Deformer maintained computational growth below 2% across three datasets while achieving substantial accuracy gains. Collectively, FGPT demonstrates stable classification performance enhancements across diverse model types at negligible computational cost, exhibiting robust generalization and augmentation effects across datasets. This confirms its efficacy and efficiency as a plug-and-play module.

Furthermore, to rigorously verify that the performance gains are not due to random variance or initialization bias, we evaluated the models across independent runs using three distinct random seeds (2023, 1949, and 100) and conducted a

comprehensive statistical significance analysis across Datasets A (Attention), B (Fatigue), and C (Mental Workload). TABLE VI presents the detailed performance metrics horizontally integrated with their corresponding statistical tests. The integration of FGPT yields consistent performance enhancements for most baseline architectures. Notably, a slight decrease in average performance is observed for EEG-ViT on Dataset A, which is primarily attributed to the fact that hyperparameters, such as the frequency token prompts, were kept fixed without fine-tuning across the three different random seeds. Nevertheless, the overall paired t-test results exhibit substantial statistical superiority. Specifically, the paired t-scores for ACC reach 5.043036, 3.485025, and 2.884428 on Datasets A, B, and C, respectively, with corresponding p-values of 0.000001, 0.000738, and 0.004185. Similarly, for the F1-macro metric, the t-scores are 4.234926, 3.280551, and 2.732828, yielding p-values of 0.000033, 0.001436, and 0.006625, respectively. Since all overall p-values across the three datasets are substantially below the standard 0.05 threshold, the enhancements brought by FGPT under these varied seed conditions are

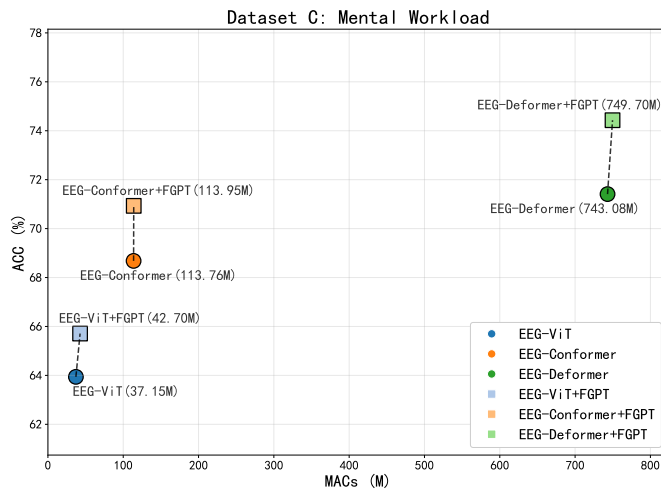


Fig. 9. Accuracy-Computation Complexity scatter plot comparison of three Transformer-Based EEG decoding models on the Mental Workload dataset before and after applying FGPT.

deemed highly statistically significant. These metrics provide strong evidence that the proposed FGPT module robustly and consistently enhances EEG decoding performance across diverse tasks, rather than causing incidental fluctuations. In terms of computational efficiency, FGPT demonstrates significant advantages. Fig. 6 illustrates that integrating FGPT incurs only minimal additional overhead across models. Taking EEG-Conformer on Dataset A as an example, FLOPs increase by merely 0.2M (0.16%), while parameters grow by 10.08K (1.54%). For EEG-ViT, parameter growth remains below 1% across three datasets. Particularly noteworthy is EEG-Deformer, which achieves a 6.53% accuracy gain on Dataset A while maintaining excellent computational efficiency, with an optimized FLOPs/parameters ratio. This characteristic is consistently observed across three datasets and different model architectures, demonstrating FGPT's exceptional versatility and stability. Overall, FGPT delivers substantial performance enhancements at a negligible resource cost, showcasing outstanding practical value.

To further validate FGPT's high efficiency, we plotted scatter diagrams comparing the ACC-MACs of different baseline models before and after applying FGPT across three datasets, where MACs denotes the cumulative accumulation operation. The results are shown in Figs. 7-9. Fig. 7 presents the comparison results on the Attention dataset. After incorporating FGPT, EEG-ViT and EEG-Conformer saw MACs increase by merely 4.4M and 0.05M respectively, whilst accuracy improved substantially as shown in TABLE I. EEG-Deformer demonstrated the most pronounced accuracy gain. Fig. 8 shows the Fatigue dataset, where applying FGPT yields substantial accuracy gains with only a marginal increase in computational complexity (MACs), representing an extremely low cost. Fig. 9 shows FGPT's outstanding performance in the Mental Workload dataset. Furthermore, post-FGPT implementation, EEG-ViT and EEG-Deformer incurred slightly higher computational costs than EEG-Conformer. This disparity relates to FGPT's sparse frequency prompt token count and dataset processing.

It should be noted that computational complexity costs are further influenced by the inherent complexity of the baseline model and the original sequence length. Nevertheless, despite these multiple factors, our FGPT method achieves highly efficient and outstanding performance.

## V. DISCUSSION

This section will further explore the important role of FGPT in EEG decoding within the Transformer framework. Fig. 10 illustrates the influence of FGPT during the attention calculation process and the representation after removing the frequency prompt token. During the attention calculation process, the frequency prompt token and the original sequence actually interact with each other, providing explicit prompt representations across different modalities. Therefore, to enhance the prompting effect of the frequency Transformer layer, we removed the frequency prompt after the attention calculation was completed and re-embedded it before each layer's attention calculation. This approach both enhances the prompting capability of the frequency token and avoids interfering with the original sequence. If the frequency token is not removed, it may implicitly prompt spatio-temporal features in subsequent networks, weakening the ability to extract frequency-domain features.

Additionally, when frequency prompts are embedded before the Transformer layer and removed after the Transformer layer, normalization layers perform computations on the concatenated sequence. Since the concatenated sequence contains information from two domains, this increases the uncertainty of normalization and raises model complexity. If concatenation embedding is performed before attention and frequency prompts are removed after attention computation, this effectively avoids computations in other layers.

Attention, as a core component of the Transformer, can process sequences of varying lengths. Compared to directly feeding the original sequence into the attention component, FGPT feeds a new sequence connected to frequency prompts into the attention method, enabling the original sequence to obtain frequency component features, as shown in Fig. 10. In Fig. 10, the frequency token participates in the entire attention calculation as a prompt, prompting the original token sequence to capture frequency information. Similarly, the original sequence serves as a spatio-temporal-domain feature to prompt the frequency-domain prompt token to capture spatio-temporal-domain components. Therefore, removing the frequency prompt token after the attention calculation reduces spatio-temporal-domain assimilation of the frequency prompt token and avoids statistical bias in the second LayerNorm. If the entire Transformer is pre-embedded with frequency prompts and the frequency prompts are removed after Transformer computation, although this method enables cross-layer frequency prompts, the frequency prompt performance of the frequency prompt tokens is assimilated by the spatio-temporal-domain features after multiple Transformer layers, reducing the frequency prompt capability. Additionally, this method cannot effectively avoid statistical calculation biases caused by distribution shifts during LayerNorm.



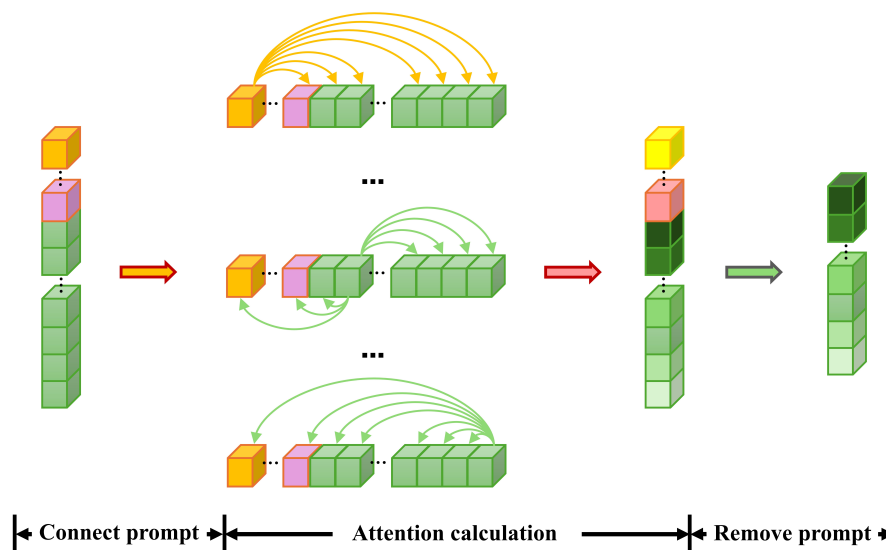


Fig. 10. The computational process for different tokens after embedding sparse frequency prompts into the attention layer, and the removal of sparse frequency prompt operations. During attention computation, sparse frequency prompt tokens prompt native sequence tokens to model frequency components, while the native sequence prompts frequency prompt tokens to complete spatio-temporal modeling.

The FGPT method for embedding and removing attention positions has another advantage: it can precisely maintain the balance between frequency-domain components and spatio-temporal-domain components. After attention calculation, the original sequence retains frequency components, while skip connections allow sequences with frequency components to enhance their perception of spatio-temporal-domain features, thereby inherently avoiding excessive frequency prompts. At this point, the learnable gating weight parameters are quantitatively adjusted through backpropagation to measure the specific contribution of frequency tokens to frequency-domain components, thereby maintaining the balance between frequency-domain and spatio-temporal-domain components in the token sequence.

In terms of model efficiency, FGPT achieves frequency prompting using sparse frequency prompt tokens. At this point, the computational complexity of attention changes from  $O(T^2D)$  to  $O((T+P)^2D)$ , where  $T$  is the original sequence length and  $P$  is the number of prompt tokens. Therefore, the computational complexity only increases slightly. Thanks to the FGPT removal mechanism, there is no increase in the computational complexity of the model in the feedforward neural network.

## VI. CONCLUSION

To mitigate the limitations in Transformer modeling caused by the absence of frequency-domain information in electroencephalograms (EEGs), this paper explores a frequency-domain prompt learning mechanism (FGPT). The proposed method introduces lightweight learnable frequency prompt tokens and employs adaptive gated fusion to integrate frequency-specific patterns. By concatenating sparse frequency prompt tokens with the original spatio-temporal token sequence and feeding them into the attention layers, the model is capable of jointly modeling long-range spatio-temporal dependencies and frequency-domain features while preserving the continuity

and intrinsic coupling of the original sequence. The prompt tokens are subsequently removed after attention computation to maintain feature integrity. Experimental results indicate that incorporating the FGPT strategy leads to improved decoding robustness. The approach demonstrates potential in enhancing the extraction of task-relevant features for EEG decoding on the evaluated datasets and exhibits certain generalization capabilities, suggesting its potential applicability to other Transformer-based EEG systems.

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